Presumed Heroes, Victims, and Villains: Narrative Framing Analysis in News Media Using Generative LLMs

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Abstract

This paper explores computational methods for framing analysis in written news media. While prior NLP research has primarily focused on topic-based classification of emphasis frames (e.g., economic, public safety), such approaches have come under scrutiny for potentially limited practical interpretability and sometimes overlooking key framing nuances.

We recast the problem as a Machine Reading Comprehension task, identifying entities portrayed as heroes, victims, and villains in newspaper articles to illuminate the narrative framing in the news stories. We present a generative language model approach that extracts these roles from news articles on climate change, sampled from outlets across the political spectrum.

Through systematic experimentation with prompting strategies and evaluation frameworks, our best configuration yields notable improvements over baseline methods, achieving 64.47% accuracy for hero identification and 66.67% for villain identification. These results suggest progress relative to earlier work, while recognizing differences in methodology and evaluation.

We also examine key limitations, including abstention behavior and evaluation complexities. Finally, we posit that computational extraction of narrative roles offers promise as a tool for media analysis, though using generative LLMs for this purpose raises significant challenges.

1 Introduction

In communication and media studies, Framing Analysis investigates how the presentation, structure, and contextualization of information influence audience perception and shape interpretation. For instance, consider the following two headlines:

- Blockades by climate activists create chaos in The Hague
- Protests in The Hague seek to bring focus to the climate crisis

Though both statements refer to the same event, a climate protest in The Hague, they project markedly different sentiments. The first frames the event as disruptive, focusing on the negative impact, while the second emphasizes the protest's intent and aligns it with a broader moral cause. These subtle linguistic choices can have significant effects on public understanding and political attitudes.

This paper explores how such framing phenomena can be studied computationally. While natural language processing (NLP) has made substantial progress in analyzing news content, much of the work in this space treats framing as a topic modeling or multi-label classification problem. These approaches often reduce complex framing choices to predefined categories or word clusters. However, recent critiques—such as those by Ali and Hassan (2022) and Otmakhova et al. (2024) —have questioned the practical interpretability and conceptual adequacy of such models. As an alternative, we focus on bringing to light the narrative framing of the message.

Our central task is to model how narratives are constructed in news reporting by identifying who or what is framed as a hero, victim, or villain. We treat this as a Machine Reading Comprehension problem and use generative large language models to extract these narrative elements from complete news articles. Through systematic experimentation with various strategies and evaluation frameworks, our optimized approach more than doubles baseline performance, comparing favorably with previous work in this domain.

We critically assess limitations of our approach, particularly those that may hinder real-world implementation. We posit that automated narrative extraction tools might support journalistic transparency initiatives, similar to those by Media Bias/Fact Check (MBFC) and AllSides, by equipping readers with a new interpretative instrument for media analysis. At the same time, we acknowledge significant challenges involved in building trustworthy and reliable systems for this purpose using generative LLMs.

2 Prior Literature

As surveyed by Ali and Hassan (2022), NLP has been extensively used for multi-label classification of frames, where they are conceptualized as themes or dimensions that can be perceived as salient in a document (e.g. economic, public safety, morality, fairness and equality). Early research relied on classical machine learning techniques, with both supervised methods like Naive Bayes classifiers and unsupervised approaches such as clustering and topic modeling. More recent advancements have leveraged transformer architectures to enhance frame detection capabilities. For example, Fremann et al. (2023) combined Longformer's capacity to process extended document contexts with traditional machine learning classifiers. Specifically addressing the data scarcity problem in the social sciences, an important development was the integration of natural language inference into framing analysis, with BERT-NLI approaches by Laurer et al. (2024) and Kuang et al. (2025).

Categorization of emphasis frames can fail to capture crucial communicative nuances. Consider that statements like *immigrants are taking our jobs* and *new wave of immigrants boost local economy* would both be classified under an "Economic" frame despite conveying opposing perspectives. Therefore, we explore a different avenue to classification in this paper.

Otmakhova et al. (2024) surveyed framing conceptualizations across media studies, cognitive linguistics, and social psychology, demonstrating that framing mechanisms remain inadequately addressed in NLP implementations. Their interdisciplinary review, alongside that of Ali and Hassan (2022), highlights a persistent gap between theoretical understandings of frames and their computational operationalization. As an alternative, Frermann et al. (2023) proposed capturing document-level narratives instead of localized emphasis frames, arguing that this approach better represents how framing functions across entire texts. We adopt this conceptual narrative approach for framing analysis.

Stammbach et al. (2022) used zero-shot classification with GPT-3 to identify archetypical narrative roles (specifically villain, hero, and victim) across diverse document types. Their method showed significant accuracy improvements over existing baselines, suggesting that large language models can effectively recognize narrative structures that traditional framing methods might miss. We build on their work and adapt it to the dataset below.

3 Data

We will use the Narrative Frame Corpus, developed by Frermann et al. (2023), available under an MIT License.

It consists of 428 manually annotated articles on climate change from the United States and the United Kingdom. The dataset is balanced across the dates of publication (2017–2019) and the four most dominant political orientation of the outlet (left, center-left, right, and questionable source) as identified by the Media Bias Fact Check website.

The authors released annotations on entities and their narrative roles (either villain, victim, or hero) in the news articles. Entities fall into one of these groups:

```
GOVERNMENTS_POLITICIANS_POLIT.ORGS
INDUSTRY_EMISSIONS
LEGISLATION_POLICIES_RESPONSES
GENERALPUBLIC
ANIMALS_NATURE_ENVIRONMENT
ENV.ORGS_ACTIVISTS
SCIENCE_EXPERTS_SCI.REPORTS
CLIMATECHANGE
GREEN_TECHNOLOGY_INNOVATION
MEDIA_JOURNALISTS
```

Some considerations apply to this dataset. First, narrative roles are not mutually exclusive; a single entity can occupy multiple roles simultaneously—for example, an individual may be both a victim and a hero within the same narrative. Second, not all roles are guaranteed to appear in every news article; certain roles, such as "hero," may be entirely absent in some cases. Third, multiple entities may fulfill the same narrative role, and the annotations may not be exhaustive; for instance, only one victim may have been identified, even if others were present. Additionally, the entity annotations require further cleaning (see more details in the Dataset Preparation section). Lastly, the dataset exhibits significant class imbalance across groups, with a notably higher number of entries for government, politicians and political organizations (774) compared to media and journalists (23).

4 Model

Our research explores computational framing analysis in written news media through the lens of narrative elements. We formulate the experiment as a Machine Reading Comprehension task: given the full text, who or what is being portrayed (framed) as a hero, victim, or villain?

Our choice of models is highly restricted by the constraints of the task and the dataset:

1. Our goal is to extract an entity, so we will need generative LLMs.

2. Small dataset. Annotating datasets for framing analysis is particularly expensive and time consuming given how subjective and nuanced the task is. In addition, news articles are generally copyrighted. Not having enough data for fine-tuning, we are left with zero and few-shot approaches.

3. Document length varies and is, as a general rule, longer than 512 tokens (these are full news articles). In this dataset, the average number of words in an article is 630, and the maximum is 1001. Therefore, we need models with long context windows.

4. Since entity annotations are not standardized and may contain typos or inconsistent formatting, the model needs to be resilient to such noise in the input data.

5. In order to run multiple experiments, the language model must be cost-effective.

4.1 Metrics

Stammbach et al. (2022) manually mapped the generated answer to one of the fixed groups available (see Table 1 for examples). This isn't a good fit for our case: it would be too time consuming as we have a substantially larger amount of articles, and the largest group (politicians or political organizations) is far too broad to illuminate a narrative angle.

Syntactic evaluation is also not adequate. Consider the following examples:

- *lack of bipartisan agreement* should not match *successful bipartisan agreement*
- *Coalition of States* should match *4 states that signed the treaty*

We attempt to match entities according to semantic similarity (in the context of the news article) using an LLM as a judge strategy. We then calculate accuracy per narrative role.

We used DSPy for prompting. Time constraints led us to choose OpenAI models, as their client works out-of-the-box with DSPy. To predict, we used GPT-4o mini. For the judge, we chose GPT-4.1 mini.

Our baseline is zero-shot prediction, drawing on findings from Stammbach et al. (2022). Though our metrics are not directly comparable, the authors conclude that GPT-3 outperforms traditional dictionary-based methods in a zero-shot setting.

5 Methods

5.1 Dataset preparation

The first step was preparing the datasets. We first cleaned the annotations as follows:

- Excluded those where the assigned entities were implausible or illogical within the narrative context (e.g., "climate change" labeled as a victim, "kids" as villains).
- Removed entities with referential ambiguity that could not be reliably interpreted (e.g., "us", "many").
- Discarded annotations that did not correspond to coherent entities, such as phrases describing events or abstract conditions rather than actors (e.g., "face temperatures of 90F 32C and higher", "for subsidizing the National Flood Insurance Program").
- Removed duplicate entries.

We created separate datasets for each narrative role (hero, victim, and villain). Each dataset contains the full text of the article and only those entities annotated with the corresponding narrative role label. For instance, the heroes dataset comprises all texts where one or more entities have been labeled as heroes, along with the relevant entity annotations.

To evaluate model generalization, we reserved a 25% hold-out set using stratified sampling based on the mbfc classification column, ensuring balanced representation across media sources using the Media Bias Fact Check (MBFC) classification. These test samples remained entirely unseen during development.

5.2 Predictor

After exploring several alternatives, our final hero prompt asks *"Does the text suggest that some entity could alleviate a problem?"* This formulation was selected because it closely parallels the question posed to human annotators, which is available in the detailed annotation instructions by Fremmann et al. (2023). It avoids direct mentions of "hero" that we found consistently biased models toward identifying individuals rather than organizations or concepts.

```
class BasicHeroSignature(dspy.
    Signature):

__doc__ = """Does the text suggest
    that some entity could alleviate
a problem?"""

text = dspy.InputField()
entity = dspy.OutputField(desc="Often
     between 1-5 words")
```

We attempted to improve performance through chain-of-thought prompting, but this approach did not yield significant improvements. We considered the advantage of explainability in making a prediction, but found that sometimes the result did not follow the reasoning, directly undermining that point.

Similarly, few-shot prediction optimization performed worse than expected, likely due to the inherent challenges of selecting representative examples given the lengthy texts, subjective nature of the task, and non-exhaustive entity lists in our dataset.

Through iterative experimentation, we discovered that explicitly identifying the central conflict in a news story before attempting to extract narrative roles improved prediction quality. This insight led us to develop a *Narrative Arc* signature that first establishes the core tension before identifying entities that play specific roles within that conflict:

```
class NarrativeArcSignature(dspy.
    Signature):

    __doc__ = """Identity narrative arc.
        Stick to the text"""

    text = dspy.InputField()
    conflict = dspy.OutputField(desc="be
        concise")
    hero = dspy.OutputField(desc="Not
        necessarily a person")
    victim = dspy.OutputField(desc="Not
        necessarily a person")
    villain = dspy.OutputField(desc="Not
        necessarily a person")
```

5.3 Judge

For evaluating prediction quality, we initially used a zero-shot approach with the prompt: *"Determine if the predicted entity is semantically equal to at least one of the gold entities."* However, this proved to be too strict a requirement: when manually going through the matches, we found many false negatives.

We achieved substantially better evaluation through a few-shot approach using the prompt: *"Determine if the predicted entity is similar in meaning to at least one of the gold entities"* and a hand-picked set of examples from the hero dev set.

To evaluate the judge, we took a sample of the results per dataset and hand-graded the matches. We concluded that the judge was correct in all but one example, which turned out to be an incorrect data point to begin with: the human annotated hero was "the wildfire", which did not make sense to us when reading the news story. In contrast, the generated entity was "firefighters fighting the wildfire". The judge matched these as equal, which is incorrect. In this outlier case, the annotated hero was wrong, the prediction was right, and the judge decided they matched.

We recognize that deciding which entities should be classified as equivalent introduces some degree of subjectivity, which we found to be inevitable. We show the examples of the few-shot judge in Table 2. All code is available on GitHub.

6 Results

Our experimental evaluation focused on several key parameters to determine the most effective approach for narrative role extraction. We specifically examined the impact of predictor type (zero-shot with or without Chain of Thought), signature choice (prompt), and judge configuration (zero or few-shot) across our datasets. We report accuracy in the following tables:

6.1 Hero dataset

6.1.1 Best signature: Narrative Arc

Judge	BasicHero	NarrativeArc
Judge zero-shot	30.26%	26.32%
Judge few-shot	57.89 %	**64.47%**

Table 3: Accuracy of zero-shot predictions on Hero dataset. Baseline in violet

Table 3 compares accuracy between predictions with our BasicHero and NarrativeArc signatures under different judge configurations. When paired with a few-shot judge, prediction with the NarrativeArc signature is more than twice as accurate as the baseline (64.47% compared to 30.26%). This suggests that incorporating the interplay of narrative roles is more effective than extracting each in isolation.

6.1.2 Chain-of-Thought prediction versus Few-Shot Evaluation

Prediction	Zero-shot Judge	Few-shot Judge
Zero-shot	26.32%	**64.47%**
Zero-shot CoT	**40.79%**	59.21%

Table 4: Accuracy of zero and few-shot Judge over zero and few-shot predictions on Hero dataset with NarrativeArc signature

Table 4 reveals a significant interaction between reasoning approach and evaluation method. When using a zero-shot judge, chain-of-thought (CoT) reasoning dramatically improves performance (from 26.32% to 40.79%). However, this advantage disappears with a few-shot judge, where CoT actually reduces accuracy from 64.47% to 59.21%.

Comparing the two columns indicates that optimizing the evaluation component to recognize semantic similarities in the context of the text yields the most improvements overall.

6.2 Few-shot judge narrative arc across datasets

Prediction	hero	villain	victim
Zero-shot	64.47%	**66.67%**	60.71%
Zero-shot CoT	63.16%	61.76%	58.33%

Table 5: Accuracy of few-shot judge with Narrative Arc across hero, villain and victim datasets

Table 5 extends our analysis across all three narrative roles (hero, villain, and victim) using the few-shot judge with NarrativeArc signature. Zero-shot prediction without CoT maintains relatively consistent performance across roles, with villain identification showing the highest success rate (66.67%), followed by hero (64.47%) and victim (60.71%). Interestingly, incorporating CoT consistently reduces performance across all three roles when paired with a few-shot judge. This suggests that for narrative role identification, the additional reasoning steps in CoT may introduce biases or complexities that make predictions less semantically aligned with human annotations.

7 Analysis

7.1 Accuracy

This is a very difficult task to begin with, and the results are better than expected. Our best model was more than twice as accurate as our baseline in the hero dataset (64.47% versus 30.26%).

While direct metric comparisons with Stammbach et al. (2022) are not one-to-one equivalent, some meaningful contrasts can be drawn. We benefited from using a newer language model while working with a larger, more complex dataset. Our data focused on broader climate change coverage versus their more narrow fracking topic, and it encompassed multiple outlets across the political spectrum (left, center-left, right, and questionable sources) from the United States and United Kingdom rather than a single newspaper in Colorado, US.

We contend our task was also more challenging: Stammbach et al. (2022) calculated accuracy based on whether predicted entities fell into 4 groups: "the public", "the government", "environmental organizations", and "the oil and gas industry". We required entity-level semantic matches with human annotations, aiming for a more nuanced illustration of narrative angles. Despite these stricter requirements, our results showed notable improvements in two roles:

- Hero identification: They achieved 50% accuracy from a 15% dictionary-based baseline, while we reached 64.47%

- Villain identification: They reported 47% accuracy from an 18% baseline, compared to our 66.67%

However, we observed lower performance for victim identification: they achieved 90% accuracy from a 65% baseline, while we reached only 60.71%. This discrepancy likely stems from their observation that the victim role typically defaults to "the public" as a category. Their category-level evaluation approach particularly benefits this more homogeneous role assignment.

There is a ceiling to achievable accuracy that would likely persist even with more powerful models or refined prompting strategies; it stems primarily from the non-exhaustive nature of ground truth annotations. Consider a news article in which a legislator advocates for a bill that address a societal problem. From a narrative perspective, both the legislator and the bill could reasonably fulfill the "hero" role (defined as entities that could alleviate a problem). However, if human annotators identified only the legislator while our model predicted the bill, this valid extraction would be counted as incorrect despite capturing the intended narrative function. Our qualitative analysis revealed numerous instances where model predictions, though in our judgment semantically valid and narratively appropriate, failed to match the specific entities in gold annotations.

An intuitive solution might involve generating all potential entities that fulfill each narrative role and defining success as the intersection with gold entities. However, this approach introduces new complexities, particularly the risk of incentivizing models to produce excessive entity candidates to maximize the chance of matching annotations, regardless of their narrative relevance.

7.2 Other considerations

Abstention challenges Our experimental design deliberately selected articles where annotators had confirmed the presence of specific narrative entities. As noted by Stammbach et al. (2022), this approach avoids the performance degradation that occurs when testing on articles lacking clear narrative roles. A significant limitation of generative language models in this and other contexts is their inability to abstain from predicting. Our experiments confirm these models almost always attempt to extract narrative roles even when none exist. For example, our baseline predictor confidently identified "yellow submarine" as the hero from the lyric *"we all live in a yellow submarine"*, a phrase with no problem-solving narrative structure.

This forced extraction tendency significantly limits its real-world applicability, as systems would require additional filtering mechanisms to prevent inappropriate role assignments in texts with incomplete or non-standard narrative structures.

We found that Chain of Thought would sometimes address this problem in short texts. Here is a test case adapted from a short story by Augusto Monterroso: with text *"When he woke up, the t-rex was there"*, a model identified the t-rex as a hero. Chain of thought, however, led it to (correctly) abstain:

```
   Prediction(
   reasoning='The text suggests a
       problem of a potentially
       dangerous situation, as a t-rex
       is a large and fearsome dinosaur.
        However, it does not indicate
       any specific entity that could
       alleviate this problem.',
   hero='None'
)
```

Developing reliable abstention capabilities remains an important direction for future work.

Bias in the LM LLMs may reinforce certain interpretations over others, subtly (or not) shaping narrative role extraction based on previous training rather than what is written in the news story. We saw one such example with an article from a rigth-leaning outlet where the gold victim was *"gas prices"*. The predicted victim was *"The global environment and future generations who will suffer the consequences of climate change if effective measures are not taken"*, which is in clear conflict with our objective. Comparison with other generative LLMs might be worthwhile.

Practical Implementation Considerations There is a growing discourse within journalism advocating a paradigmatic shift from the traditional ideal of objectivity toward a model grounded in transparency. Some scholars and practitioners question the feasibility and desirability of journalistic objectivity, arguing that the claim to a neutral, detached stance often obscures the subjective choices inherent in reporting practices. In contrast, transparency is proposed as a more honest and ethically responsive framework, one that acknowledges the stance of the journalist and the interpretative nature of news production. Proponents contend that transparency does not undermine credibility; rather, it enhances trust by openly disclosing the methods, sources, and limitations involved in reporting.

While tools identifying narrative framing could potentially help with transparency, the trustworthiness of our current method remains questionable for real-world deployment. The system's inability to reliably abstain from extraction and its imperfect accuracy even under favorable conditions raise concerns about dependability. While our approach offers some degree of explainability as users can examine model predictions and potentially request explanations, our evaluation methodology (and probably any that is scalable to a real-world application) relies on language models to judge the outputs of other language models, essentially asking LLMs to grade their own homework. Although LLM as a judge has many applications in industry, it could undermine confidence in a system whose goal is increased transparency. This could be alleviated by a human-in-the-loop evaluation framework and open-sourcing the code.

8 Conclusion

In this paper, we explored computational approaches to narrative framing analysis by identifying heroes, victims, and villains in news articles about climate change. By recasting framing analysis as a Machine Reading Comprehension task rather than relying on traditional topic-based classification, we addressed limitations in prior work that sometimes failed to capture or explain framing nuances.

Our generative language model approach demonstrated promising results, achieving 64.47% accuracy for hero identification and 66.67% for villain identification when using a narrative-aware extraction zero-shot prediction strategy combined with a few shot LLM judge that semantically matched predicted entities to human annotated ones.

These results suggest meaningful progress in computational narrative framing analysis, though direct comparisons with previous work must acknowledge methodological differences.

Key challenges remain before such systems could be deployed in practical media analysis contexts.

Known Limitations

Reliability and trustworthiness for real-world implementation. We explored these limitations in depth in the Analysis section:

Performance metrics, while encouraging, remain modest in absolute terms. The difficulty in establishing standardized comparison metrics across different papers further complicates meaningful evaluation of progress in this domain.

Current generative models consistently struggle with abstention- recognizing when narrative roles are genuinely absent rather than forcing extraction where inappropriate.

The LLM as a judge strategy introduces some degree of subjectivity by the person or people implementing it.

Language model biases present particular concerns in narrative role analysis, potentially reinforcing harmful stereotypes. These could manifest as systematic tendencies to assign less powerful roles to feminine-coded entities or overrepresenting certain demographic groups in villain roles, among other potential harms.

Cost, monetary and environmental. We deliberately chose very cost-effective models resulting in a total expenditure of under 3 Euros, despite the length of documents and the number of experiments conducted. Nonetheless, we acknowledge the steep environmental cost associated with the use of large general-purpose models. Due to time constraints, alternative models were not explored in this study. Future work may consider the use of smaller pre-trained QA models. For instance, Stammbach et al. (2022) report only a modest reduction in accuracy when using the UnifiedQA-T5-large model available on Hugging Face. The length of the news articles in this dataset limited our choices, but alternatives like first identifying the conflict and then extracting the narrative roles with a smaller model is worth exploring.

Authorship Statement

The author confirms sole responsibility for the following: study conception and design, modeling, coding, analysis and interpretation of results.

Claude and ChatGPT were used as writing and editing assistants for parts of the manuscript. All generated text was critically reviewed and adjusted by the author as needed.

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A Appendix: tables

Generated Answer	Gold Group	Match
the citizens of Boulder County	General or specific public	Yes
The City Council	Environmental Orgs	No

Table 1: Example table showing generated answers, gold groups, and how they were matched in (Stammbach et al., 2022)

Generated Answer	Gold Entities	Match
Environmental safety regulations and advocates for responsible drilling practices	[Offshore drilling safety regulations]	Yes
California and the coalition of states fighting for environmental protection.	[California lawsuit]	Yes
The local community and concerned citizens who actively protested against the plant.	[Community protest, the local press]	Yes
Justin Trudeau	[Political leaders]	No
Peter Kalmus, as a climate scientist and advocate for personal action against climate change	[some ordinary Americans, climate action]	No
Julian Castro	[Julian Castro's Climate Plan]	Yes
Rep. Alexandria Ocasio-Cortez	[Green New Deal]	No

Table 2: Examples in EntitiesMatchFewShot judge